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V12M

Virudhunagar District
Common Quarterly Examination - 2025

Standard 12
MATHEMATICS
PART - I

Time: 3.00 Hours Marks: 90

Choose the correct answer: **20x1=20**

1) If $A^T A^{-1}$ is symmetric, then A^2 =
   a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$

2) If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ then adj(adj A) is
   a) $\begin{bmatrix} 3 & -3 & -4 \\ 2 & -3 & -4 \\ 0 & -1 & -1 \end{bmatrix}$ b) $\begin{bmatrix} 6 & -6 & 8 \\ 4 & -6 & 8 \\ 0 & -2 & 2 \end{bmatrix}$ c) $\begin{bmatrix} -3 & 3 & -4 \\ -2 & 3 & -4 \\ 0 & 1 & -1 \end{bmatrix}$ d) $\begin{bmatrix} 3 & -3 & 4 \\ 0 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$

3) If A is a non-singular matrix of order n with rows and columns are equal, then adj(adj A) =
   a) $|A|^{n-1}A$ b) $|A|^n A$ c) $|A|^{n-2} A$ d) $|A|^{n-3} A$

4) The area of the triangle formed by the complex numbers z, iz and z + iz in the Argand's diagram is
   a) $\frac{1}{2}|z|^2$ b) $|z|^2$ c) $\frac{3}{2}|z|^2$ d) $2|z|^2$

5) If $w \neq 1$ is a cubic root of unity and $(1+\omega)^7 = A + B\omega$, then (A, B) equals
   a) (1, 0) b) (−1, 1) c) (0, 1) d) (1, 1)

6) If the amplitude of a complex number is $\frac{\pi}{2}$ then the number is
   a) purely imaginary b) purely real
   c) 0 d) neither real nor imaginary

7) A zero of $x^3 + 64$ is
   a) 0 b) 4 c) 4i d) −4

8) Every polynomial equation of degree $n \geq 1$ has ..... root in C
   a) atmost one root b) atleast one root
   c) atleast 2 roots d) atmost 2 roots

9) $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to
   a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ d) $\tan^{-1}\left(\frac{1}{2}\right)$

10) If $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$, then the value of x is
   a) 4 b) 5 c) 2 d) 3

11) If $\cot^{-1}\sqrt{\sin\alpha} + \tan^{-1}\sqrt{\sin\alpha} = u$, then cos 2u is equal to
   a) $\tan^2\alpha$ b) 0 c) −1 d) tan 2

12) The length of the diameter of the circle which touches the x-axis at the point (1, 0) and passes through the point (2, 3)
   a) $\frac{6}{5}$ b) $\frac{5}{3}$ c) $\frac{10}{3}$ d) $\frac{3}{5}$

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13) If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is

a) 3 b) −1 c) 1 d) 9

14) The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ having centre at (0, 3) is

a) $x^2+y^2-6y-7=0$ b) $x^2+y^2-6y+7=0$ c) $x^2+y^2-6y-5=0$ d) $x^2+y^2-6y+5=0$

15) The locus of the point of intersection of perpendicular tangents to the parabola $y^2 = 4ax$ is

a) $x = a$ b) $y = a$ c) $x = -a$ d) $y = -a$

16) If a and b are parallel vectors, then $[a, b, c]$ is equal to

a) 2 b) −1 c) 1 d) 0

17) If a, b, c are non-coplanar, non -zero vectors such that $[a, b, c] = 3$, then $([a\times b, b\times c, c\times a])^2$ is equal to

a) 81 b) 9 c) 27 d) 18

18) If the length of the perpendicular from the origin to the plane $2x+3y+\lambda z=1$, $\lambda>0$ is $1/5$, then the value of λ is

a) $2\sqrt{3}$ b) $3\sqrt{2}$ c) 0 d) 1

19) The coordinates of the point where the line $r = (6i - j - 3k) + t(-i + 4k)$ meets the plane $r.(i + j - k) = 3$ are

a) (2, 1, 0) b) (7, −1, −7) c) (1, 2, −6) d) (5, −1, 1)

20) Equivalent matrices are obtained by

a) taking inverses b) taking transposes

c) taking adjoints d) taking finite number of elementary transformations

PART - II

Answer any 7 of the following: Q.No. 30 is compulsory. **7x2=14**

21) If adj (A) = $\begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$ find A^{-1}

22) Simplify: $\sum_{n=1}^{102} i^n$

23) Find the modulus and principal argument of $\sqrt{3} - i$

24) Find the value of $\sin^{-1}\left(\sin\left(\frac{5\pi}{4}\right)\right)$

25) Determine whether $x + y - 1 = 0$ is the equation of a diameter $x^2+y^2-6x+4y+C=0$ for all possible values of C.

26) If $a = i - 2j + 3k$, $b = 2i + j - 2k$, $c = 3i + 2j + k$ find $(a\times b)\times c$

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27) Find the distance between the parallel planes $x+2y-2z+1=0$ and $2x+4y-4z+5=0$

28) Solve $2x-y=7$ and $3x-2y=11$ by matrix inversion method.

29) If α and β are the roots of the quadratic equation $2x^2-7x+13=0$, construct a quadratic equation whose roots are α^2 and β^2

30) Find the latus rectum of the parabola $y^2 = 4ax$

PART - III

Answer any 7 of the following: Q.No. 40 is compulsory. **7x3=21**

31) Solve the equations $5x-2y+16=0$, $x+3y-7=0$ by Cramer's rule

32) Find the rank of the matrix $\begin{bmatrix} 4 & 3 & 1 & -2 \\ -3 & -1 & -2 & 4 \\ 6 & 7 & -1 & 2 \end{bmatrix}$

33) Simplify : $\left(\sin\frac{\pi}{6}+i\cos\frac{\pi}{6}\right)^{18}$

34) Show that the equation $x^9-5x^5+4x^4+2x^2+1=0$ has atleast 6 imaginary solutions.

35) Show that $\cot\left(\sin^{-1}x\right)=\frac{\sqrt{1-x^2}}{x}$, $-1\le x\le 1$ and $x\neq 0$

36) Find the equation of the hyperbola with foci $(\pm 2, 0)$, eccentricity $e = \frac{3}{2}$

37) A room 34m long is constructed to be a whispering gallery. The room has an elliptical ceiling. If the maximum height of the ceiling is 8m, determine where the foci are located

38) Find the vector equation in parametric form and cartesian equations of the line passing through $(-4, 2, -3)$ and as parallel to the line $\frac{-x-2}{4}=\frac{y+3}{-2}=\frac{2z-6}{3}$

39) If $a=i-j$, $b=j-k$, $c=k-i$ then find the value of $\left[a-b, b-c, c-a\right]$

40) Show that the points $1, -\frac{1}{2}+i\frac{\sqrt{3}}{2}$ and $-\frac{1}{2}-i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle

PART - IV

Answer all the questions. **7x5=35**

41) a) If $A=\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ then verify that $A(\text{adj } A) = (\text{adj } A)A = |A|\, I_3$

(OR)

b) Find the vertex, focus, equation of directrix and length of latus rectum

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of the parabola $y^2 - 4y - 8x + 12 = 0$

42) a) Suppose z_1, z_2 and z_3 are the vertices of an equilateral triangle inscribed in the cricle $|z| = 2$. If $z_1 = 1+i\sqrt{3}$, then find z_2 and z_3

(OR)

b) Find the parametric form of vector equation and cartesian equations of the plane passing through the points (2, 2, 1), (1, −2, 3) and parallel to the straight line passing through the points (2, 1, −3) and (−1, 5, −8)

43) a) If $z = x+iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, then prove that $x^2+y^2=1$

(OR)

b) Solve the equation $x^3-9x^2+14x+24=0$ if it is given that two of its roots are in the ratio 3 : 2

44) a) If $\cos^{-1}x + \cos^{-1}y+\cos^{-1}z=\pi$ and $0 < x, y, z < 1$, show that $x^2+y^2+z^2+2xyz = 1$

(OR)

b) Prove that $\tan^{-1} x + \tan^{-1}\frac{2x}{1-x^2} = \tan^{-1}\frac{3x-x^3}{1-3x^2}$ $|x| < \frac{1}{\sqrt{3}}$

45) a) Solve : $(x-5)(x-7)(x+6)(x+4) = 504$

(OR)

b) Prove that the point of intersection of the tangents at t_1 and t_2 on the parabola $y^2=4ax$ is $[at_1 t_2, a(t_1+t_2)]$

46) a) Prove that $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ by vector method

(OR)

b) For what values of k, the system of linear equations $kx+y+z=1$, $x+ky+z=1$, $x+y+kz=1$ have i) unique solution ii) infinitely many solution iii) no solution

47) a) If $a = i+j+k$, $b = 2i+k$, $c = 2i+j+k$, $d = i+j+2k$, then verify that $(a\times b)\times(c\times d) = [a,b,d]c - [a,b,c]d$

(OR)

b) A bridge has a parabolic arch that is 10 m high in the centre and 30 m wide at the bottom. Find the height of the arch 6m from the centre on either sides.

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